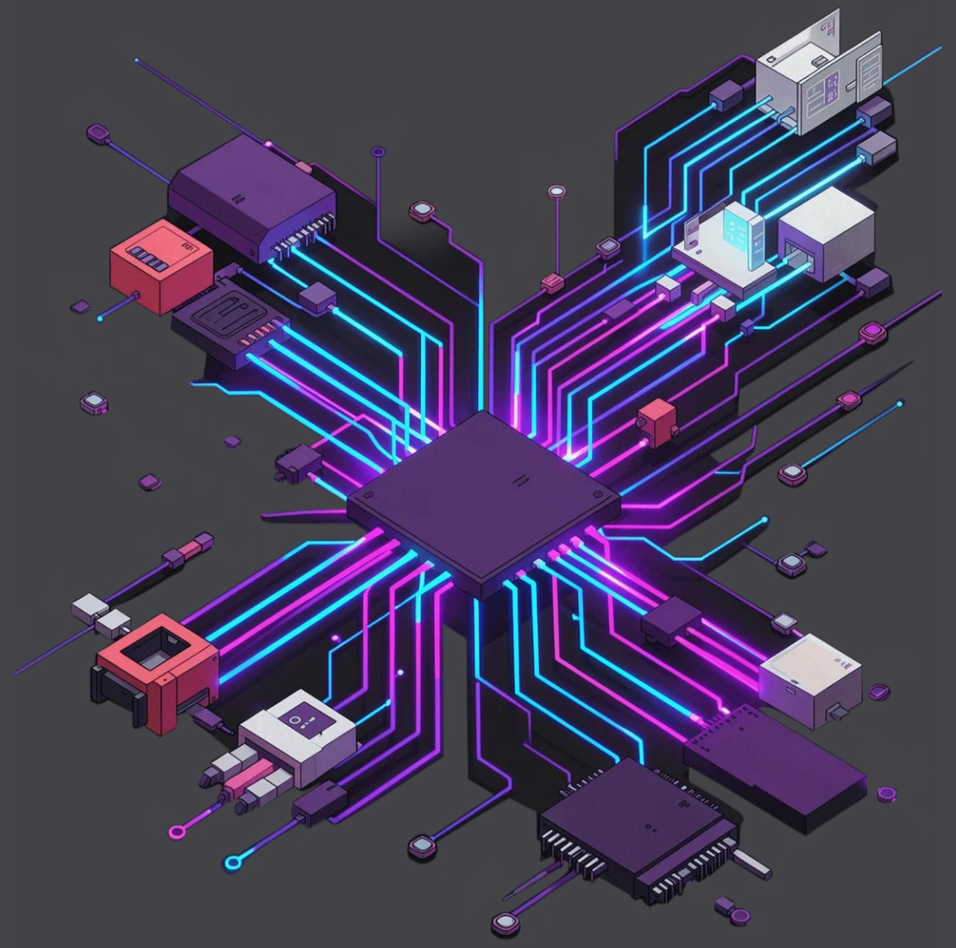
SARANSH
(2712)



AI Automation

Transforming Manual Work into Smart Systems

- Tasks performed automatically using intelligent systems
- Combines AI, workflows, and integrations
- Faster, smarter, and error-free operations

> Work smarter, not harder with automation

Group 3

Saransh	2520982712
Tripti	2520982746
Khushboo	2520982615
Nakshatra	2520982654

SARANSH
(2712)

THE CHALLENGE

The Problem

📄 Manual Expense Tracking

Employees spend hours on repetitive data entry and paper-based processes.

⏳ Time-Consuming Approvals

Approval cycles drag on for days, delaying reimbursements and decisions.

❌ Data Duplication & Errors

Human errors in spreadsheets lead to costly reconciliation issues.

📉 No Real-Time Visibility

Leadership lacks instant access to spend data for informed decisions.

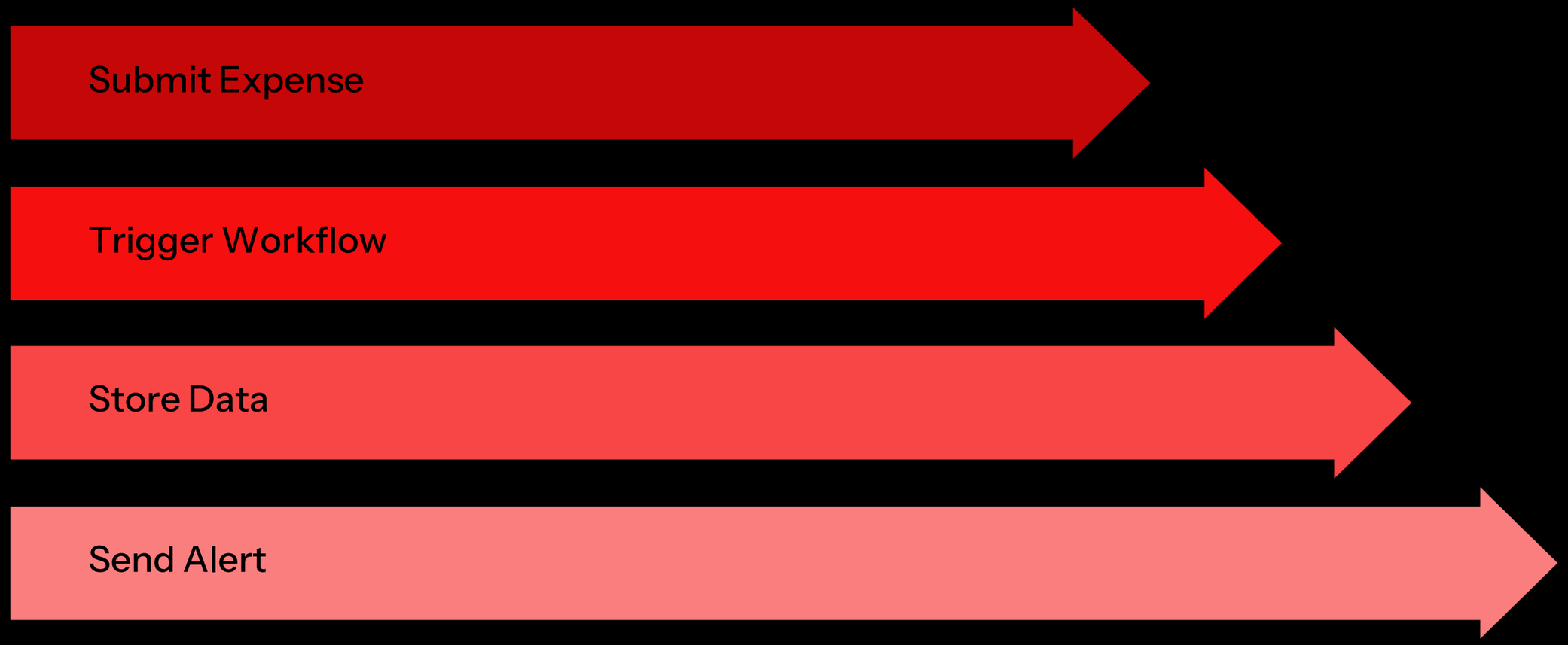
> Traditional systems are slow, inefficient, and error-prone

SARANSH
(2712)

OUR SOLUTION

AI-Based Expense Automation System

A fully integrated, end-to-end automation pipeline that eliminates manual touchpoints across the entire expense lifecycle.



One integrated system. Fully automated workflow — from submission to insight in seconds.

NAKSHATRA
(2654)

TECHNOLOGY STACK

Tools & Technologies Used

n8n

Open-source workflow automation engine that orchestrates the entire pipeline.

Google Forms

Frictionless data collection interface for employee expense submissions.

Google Sheets

Centralized, real-time data storage with automatic row-level updates.

Email Integration

Instant automated notifications to approvers and finance teams.

Power BI

Interactive dashboards delivering live spend analytics and insights.

KHUSHBOO
(2615)

SYSTEM WORKFLOW

How It Works

01

Employee Submits Expense

Fills out a Google Form — simple, guided, and error-resistant.

02

Data Captured Instantly

Form response is recorded and structured automatically.

03

n8n Triggers Automation

Workflow engine activates, validating and routing the data.

04

Data Stored in Google Sheets

Expense record appended to the master sheet in real time.

05

Email Notification Sent

Approvers and stakeholders receive instant automated alerts.

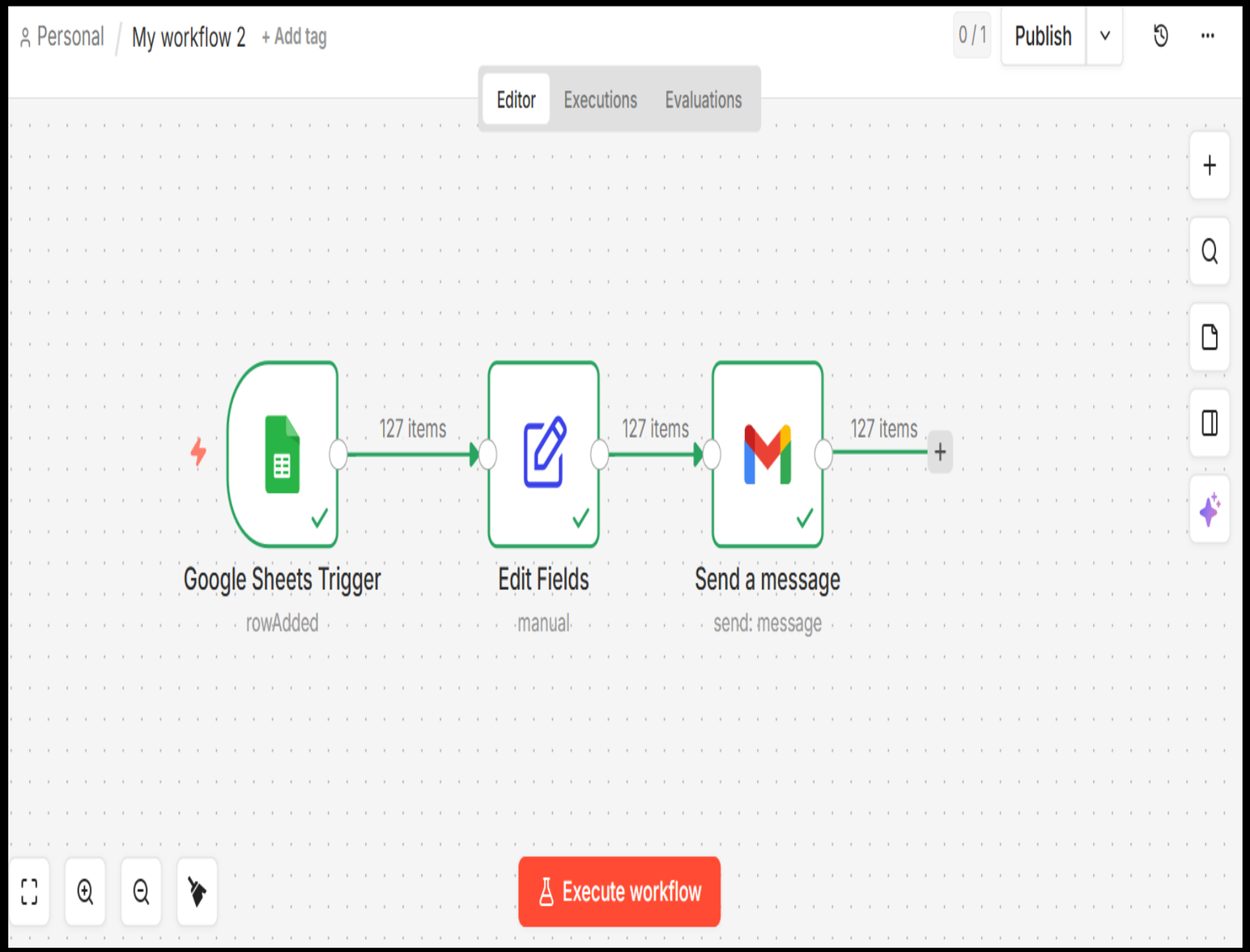
06

Dashboard Updates in Power BI

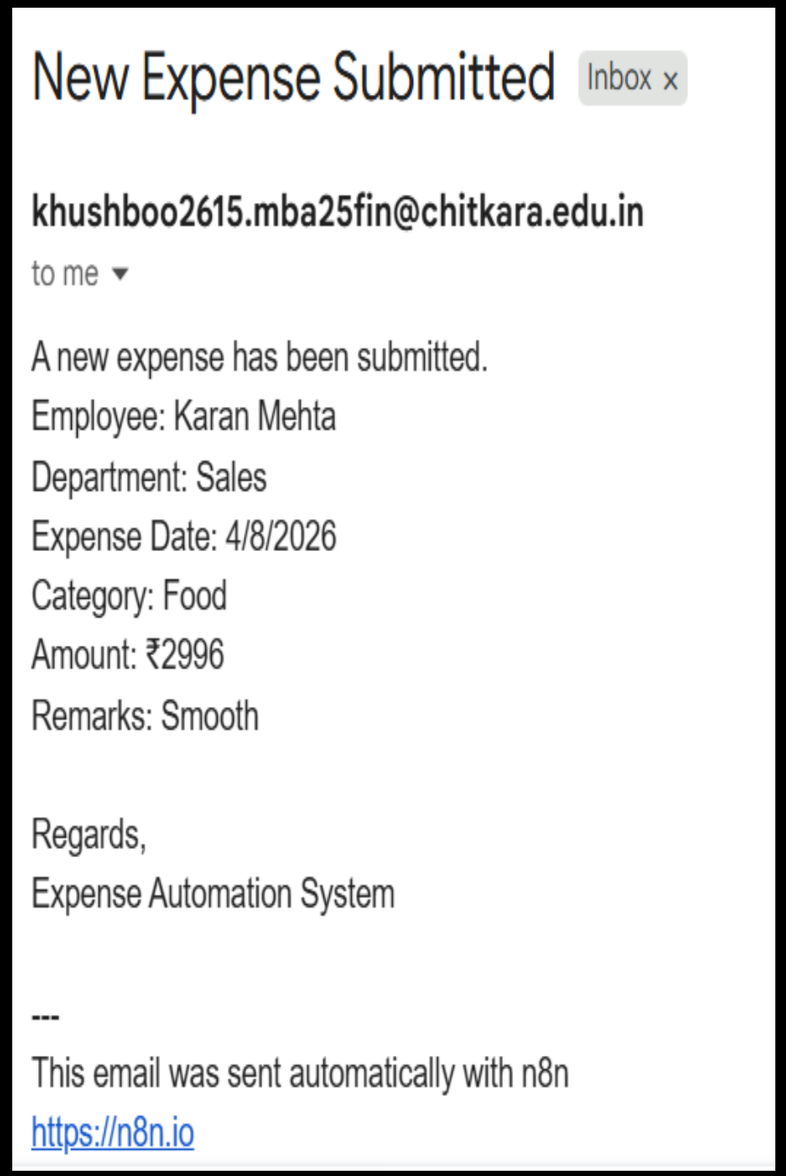
Live analytics refresh automatically — no manual intervention.

KHUSHBOO
(2615)

WORKFLOW



OUTPUT



New Expense Submitted Inbox x

khushboo2615.mba25fin@chitkara.edu.in

to me

A new expense has been submitted.
Employee: Karan Mehta
Department: Sales
Expense Date: 4/8/2026
Category: Food
Amount: ₹2996
Remarks: Smooth

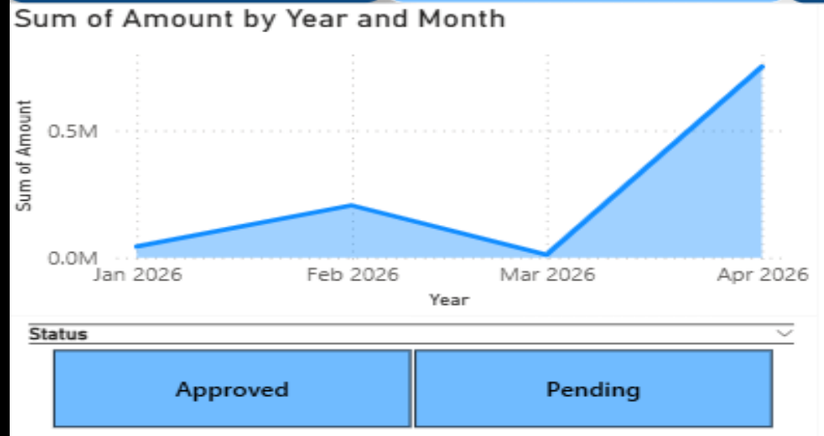
Regards,
Expense Automation System

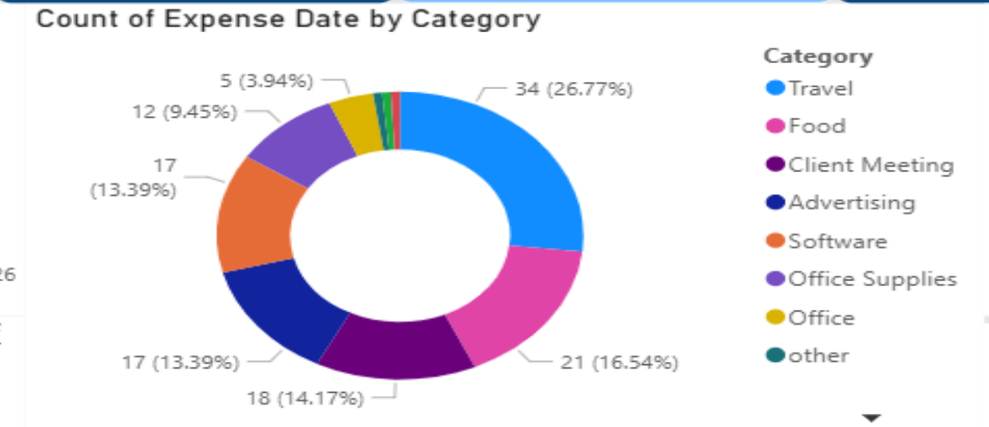
This email was sent automatically with n8n
https://n8n.io

KHUSHBOO
(2615)

POWER BI DASHBOARD

1M	127	7.94K	1M	200K	127
Sum of Amount	Count of Employee Name	average expense	total expense	highest expenses	claims count





Department

- finance
- healthcare
- HR
- IT
- marketing
- Sales

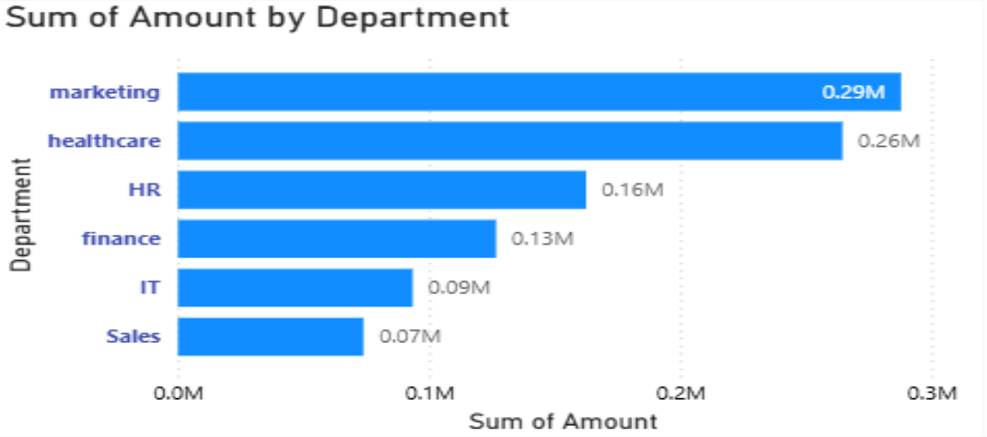
Category: All

Advertising	Office	Software
Client Meeting	Office Supplies	Travel
Food	other	Travel, Office

Expense Date

08-01-2026 30-04-2026

Employee Name	Department	Sum of Amount	Status
Aarav Sharma	finance	4084	Pending
Aarav Sharma	IT	6252	Approved
Aarav Sharma	marketing	5067	Pending
Aarav Sharma	Sales	4904	Approved
Aarav Sharma	Sales	2509	Pending
Aditya Singh	HR	5110	Approved
Aditya Singh	IT	2347	Pending
Ajay	marketing	50000	Approved
Akshu	HR	5100	Approved
Total		**1008833**	



NAKSHATRA
(2654)

KEY BENEFITS

What You Gain

Saves Time

Eliminates hours of manual data entry every week.

Reduces Errors

Automated validation removes human input mistakes.

Real-Time Tracking

Live dashboards keep every stakeholder informed instantly.

Instant Notifications

Automated alerts ensure no approval is ever delayed.

Better Decisions

Data-driven insights empower smarter financial choices.

Up to 80% reduction in manual effort — teams reclaim time for high-value work.

CONCLUSION

What We Achieved

Successfully Automated Expense Tracking

End-to-end pipeline replaces manual processes entirely.

Reduced Manual Work & Human Errors

Automation handles validation, storage, and routing.

Improved Efficiency & Accuracy

Consistent, reliable data flow across every step.

Enabled Real-Time Insights

Leadership gains instant visibility into spend and trends.

INDUSTRY USE

TRIPTI (2746)

- Finance – Invoice & Billing
- E – Commerce – Order processing automation
- Healthcare – Patient data

Source : https://versich.com/blog/n8n-workflow-automation-use-cases-10-industries-10-real-solutions-2026/?utm_source=chatgpt.com https://versich.com/blog/n8n-workflow-automation-use-cases-10-industries-10-real-solutions-2026/?utm_source=chatgpt.com

How Companies Use This Today

Time saving – 80% to 95% (approx.)

Cost reduction = 30%- 60% (approx.)

Accuracy improvement = 95%-99% (approx.)

> Companies are shifting towards fully automated, data-driven operations

TRIPTI
(2746)

FUTURE SCOPE

What's Next

1

AI-Based Fraud Detection

ML models flag anomalies and suspicious patterns in real time.

2

Smart Approval Systems

AI-driven routing and auto-approvals based on policy rules.

3

Mobile App Integration

On-the-go expense capture with camera OCR and push alerts.

4

Predictive Analytics

Forecast spend trends and proactively control costs.

5

ERP Integration (e.g., SAP)

Seamless sync with enterprise resource planning systems.

TRIPTI
(2746)

The Future of Work is Automated

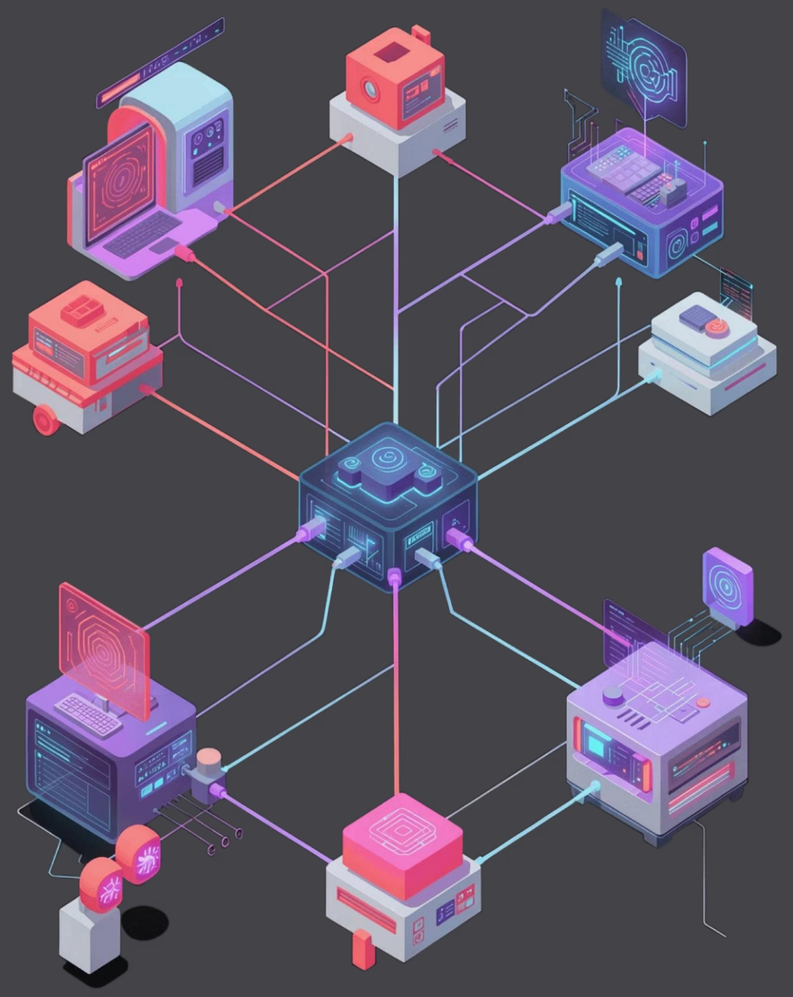
AI-powered expense automation is no longer a luxury — it's a competitive necessity. Organizations that automate today are building the foundation for smarter, faster, and more resilient operations tomorrow.

Automate

Replace manual tasks with intelligent workflows

Integrate

Connect your tools for seamless data flow

Scale

Grow without adding operational complexity